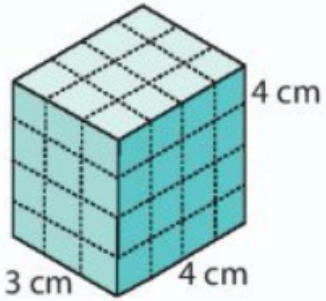
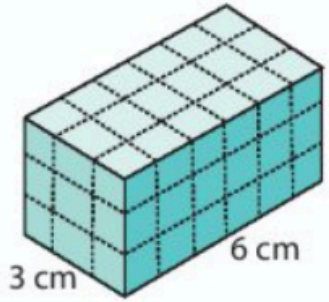
Exercise 5E

1. How many 1 cm^3 solids are in each of the diagrams over.

(b)

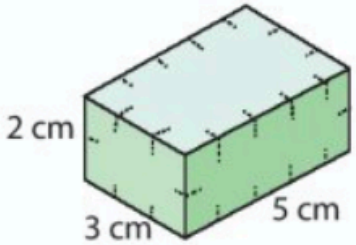


(c)

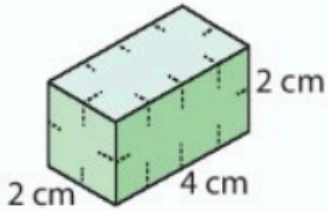


2. Find the volume of each of the rectangular solids shown:

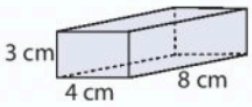
(i)



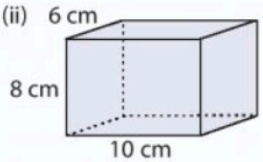
(ii)



3. Find the volume of each of the following solids: (i)

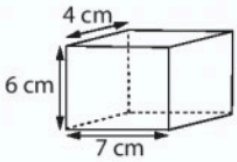


(ii)

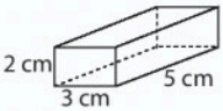


4. Find the volume, in cm^3, of each of the following blocks:

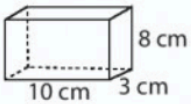
(i)



(ii)

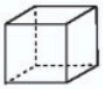


(iii)

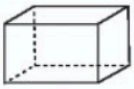


5. Look at these diagrams of 3-dimensional shapes.
Dotted lines are used to show the edges which cannot be seen when you look at the shape from one side.

(a)

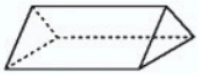


(b)



(c)

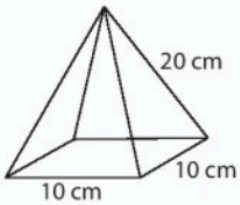
(d)



Copy and complete this table:

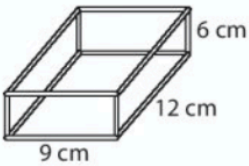
	Number of faces	Number of vertices	Number of edges
a: Cube			
b: Cuboid			
c: Pyramid			
d: Prism			

6. The diagram shows a pyramid.
A model of the pyramid is to be made using straws.
The straws are each 10 cm long and are joined using pipe cleaners.

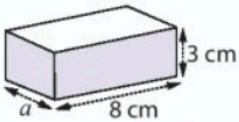


(a) How many edges does the pyramid have?
(b) How many vertices does the pyramid have?
(c) How many straws are needed to make the pyramid?
(d) What is the total length of the edges of the pyramid?

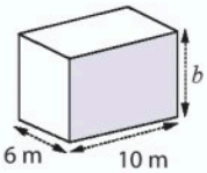
7. On the right is the skeleton of a rectangular box made of straws.
Calculate the total length of straw necessary to make the skeleton.



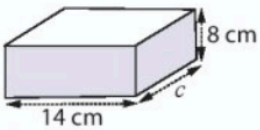
8. The volume of each of the following rectangular solids is given.
Find the length of the side marked with a letter.



Volume = 120 cm^3



Volume = 420 cm^3



Volume = 2240 cm^3

9. A rectangular block is 14 cm long and 6 cm wide.
If its volume is 672 cm^3, find its height.

10. A packet of pastry has a volume 300 cm^3.
How thick will it be if it is rolled out to form a rectangle:
(i) 15 cm × 10 cm (ii) 25 cm × 24 cm?